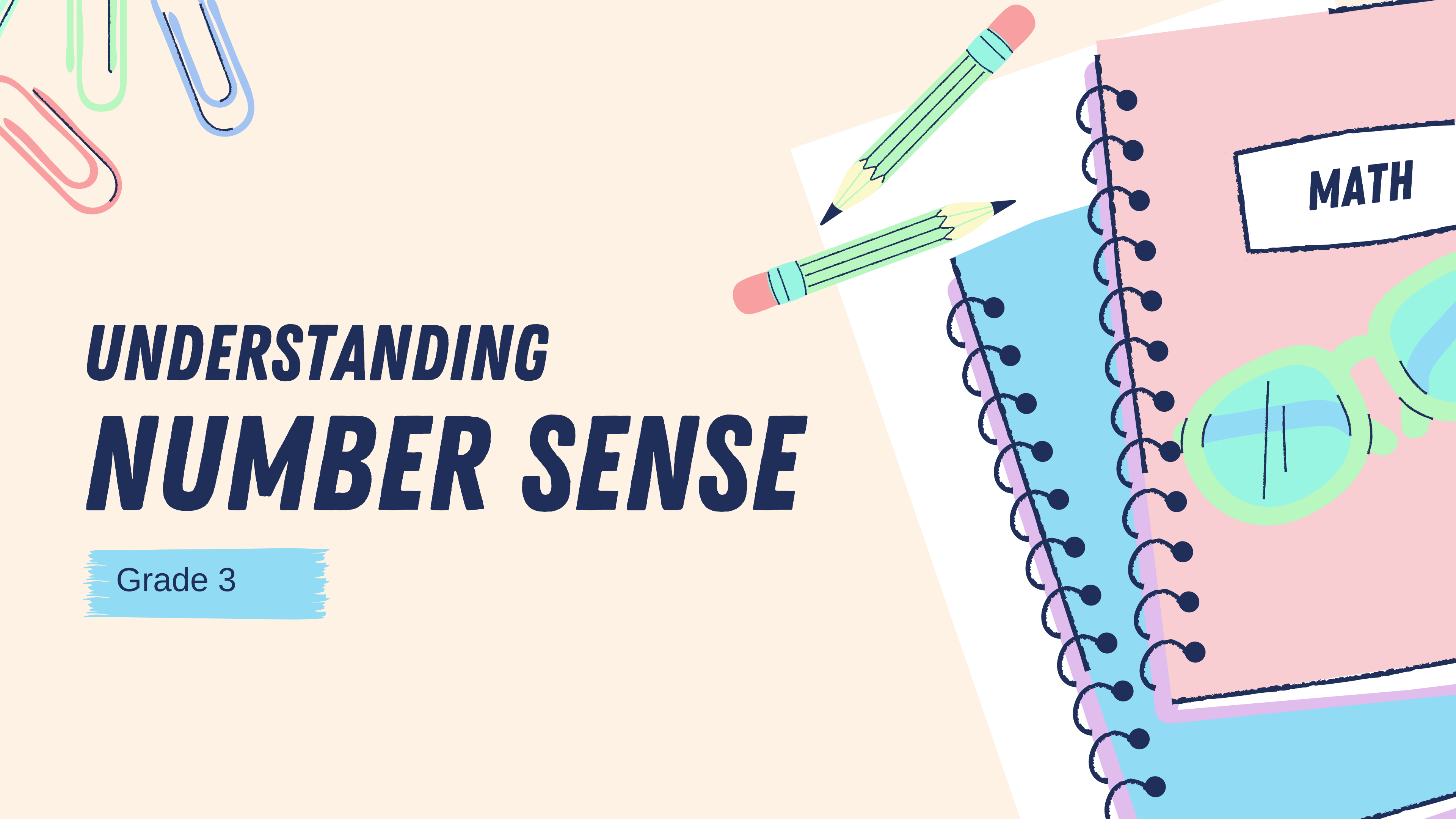


UNDERSTANDING NUMBER SENSE

Grade 3



WHAT IS PLACE VALUE ?

Place value helps us understand the value of each digit in a number.

1	3	4	6	5	2	8
MILLIONS	HUNDRED THOUSANDS	TEN THOUSANDS	THOUSANDS	HUNDREDS	TENS	ONES

PLACE VALUE VOCABULARY

DIGIT A symbol that represents a whole number.

3

VALUE The amount each digit is worth.

3000

THOUSANDS

3

HUNDREDS

4

TENS

6

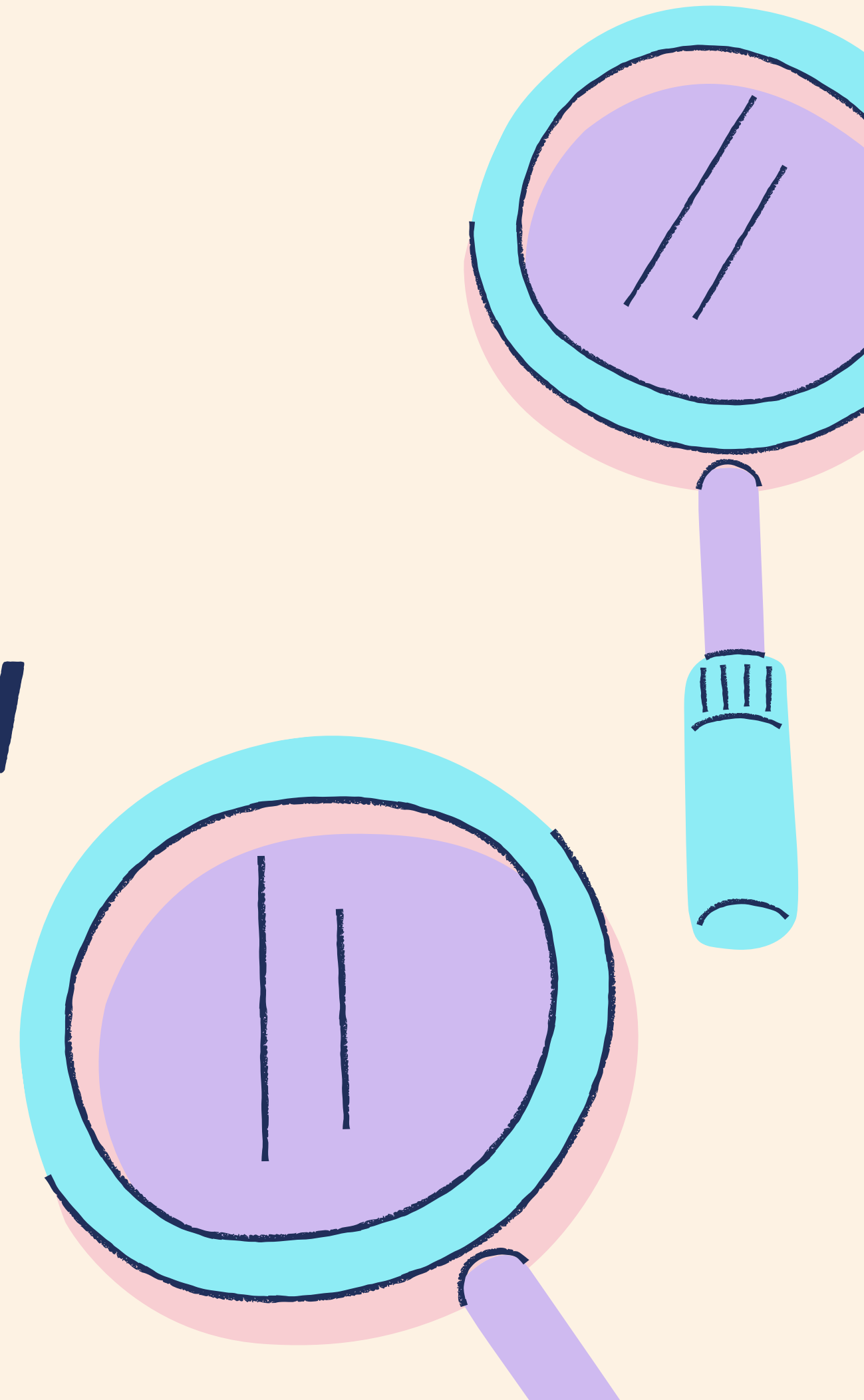
ONES

1

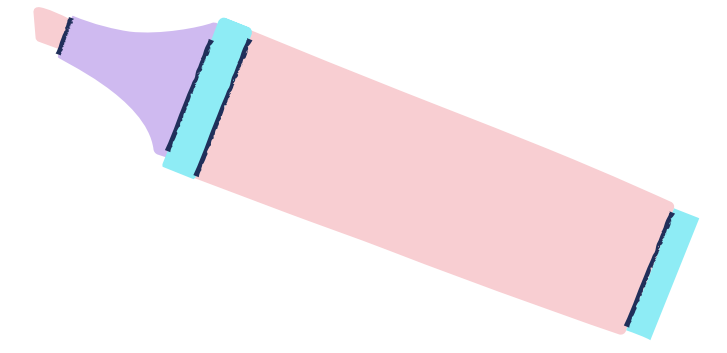


,

***NOW, LET'S DELVE DEEPER
INTO THE REPRESENTATION
OF NUMBERS.***



UNDERSTANDING STANDARD FORM



When whole numbers are written using the digits 0 to 9, they are referred to as being in standard form. In this form, the value of each digit is determined by its position.

THOUSANDS	HUNDREDS	TENS	ONES
3	4	6	1

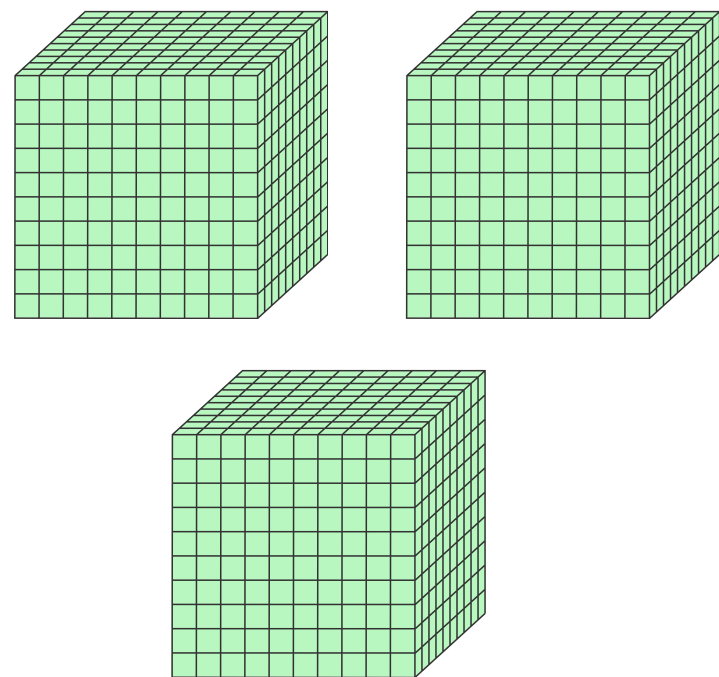
Standard Form 3,461

EXPLORING PLACE VALUE BLOCKS

Base ten blocks are a useful tool for representing numbers visually.

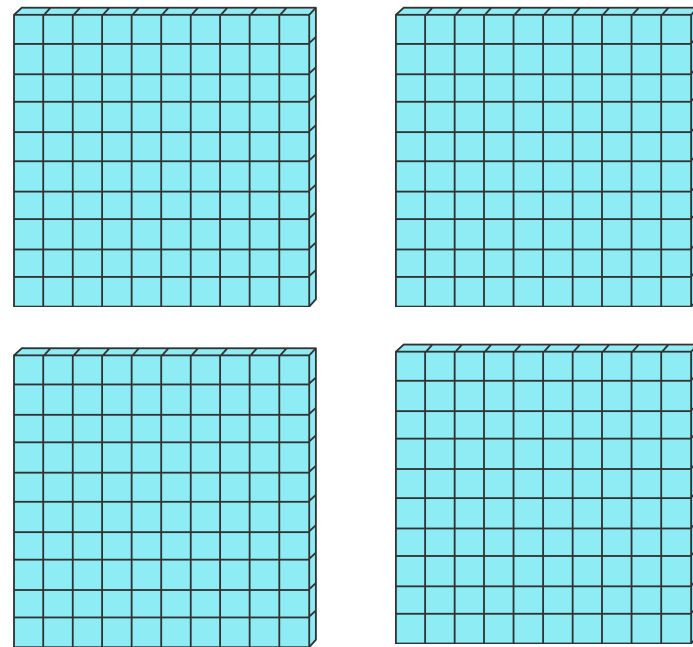
THOUSANDS

3



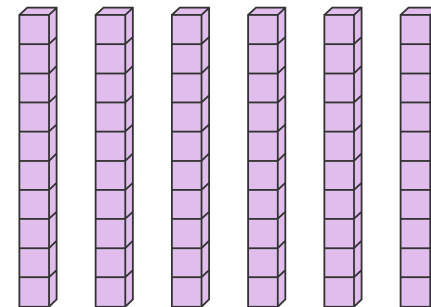
HUNDREDS

4



TENS

6

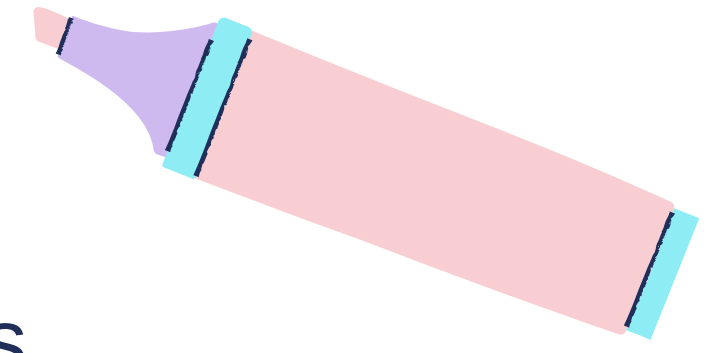


ONES

1



UNDERSTANDING WORD FORM



Word form refers to the written representation of numbers using words, rather than numerals.

THOUSANDS

3

HUNDREDS

4

TENS

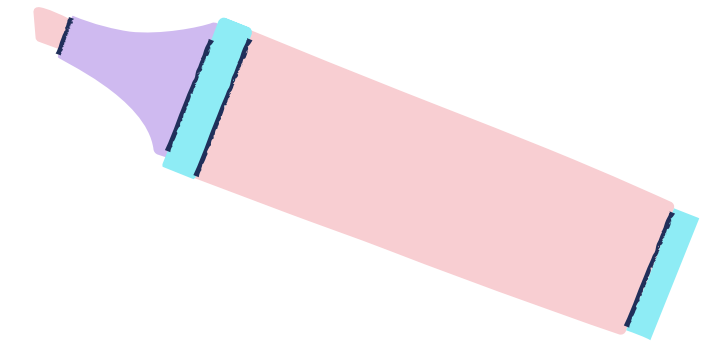
6

ONES

1

Word Form Three thousand, four hundred sixty-one

UNDERSTANDING EXPANDED FORM



Expanded form is a method of representing a number by adding the place values of its digits.

THOUSANDS	HUNDREDS	TENS	ONES
3	4	6	1

EXPANED FORM $3000 + 400 + 60 + 1$

PLACE VALUE TABLE

THOUSANDS	HUNDREDS	TENS	ONES
4	5	6	7

Use the number in the top right corner to fill out the place value table and accurately represent numerical values.

STANDARD FORM

WORD FORM

EXPANDED FORM

BASE TEN BLOCKS

PLACE VALUE TABLE

THOUSANDS	HUNDREDS	TENS	ONES
4	5	6	7

Use the number in the top right corner to fill out the place value table and accurately represent numerical values.

STANDARD FORM

4,567

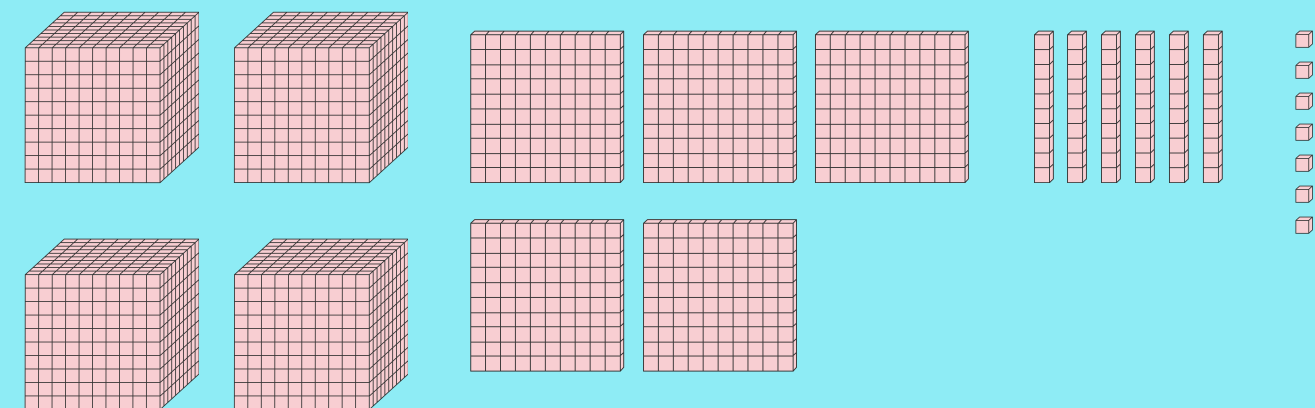
WORD FORM

Four thousand, five hundred sixty-seven.

EXPANDED FORM

$4000 + 500 + 60 + 7$

BASE TEN BLOCKS



ANSWER

PLACE VALUE TABLE

THOUSANDS	HUNDREDS	TENS	ONES
2	3	1	2

Use the number in the top right corner to fill out the place value table and accurately represent numerical values.

STANDARD FORM

WORD FORM

EXPANDED FORM

BASE TEN BLOCKS

PLACE VALUE TABLE

THOUSANDS	HUNDREDS	TENS	ONES
2	3	1	2

Use the number in the top right corner to fill out the place value table and accurately represent numerical values.

STANDARD FORM

2,312

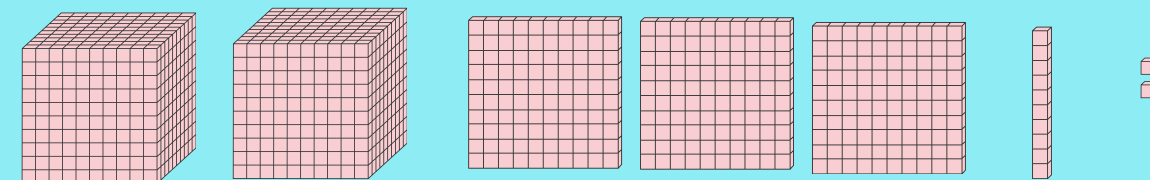
WORD FORM

Two thousand, three hundred twelve.

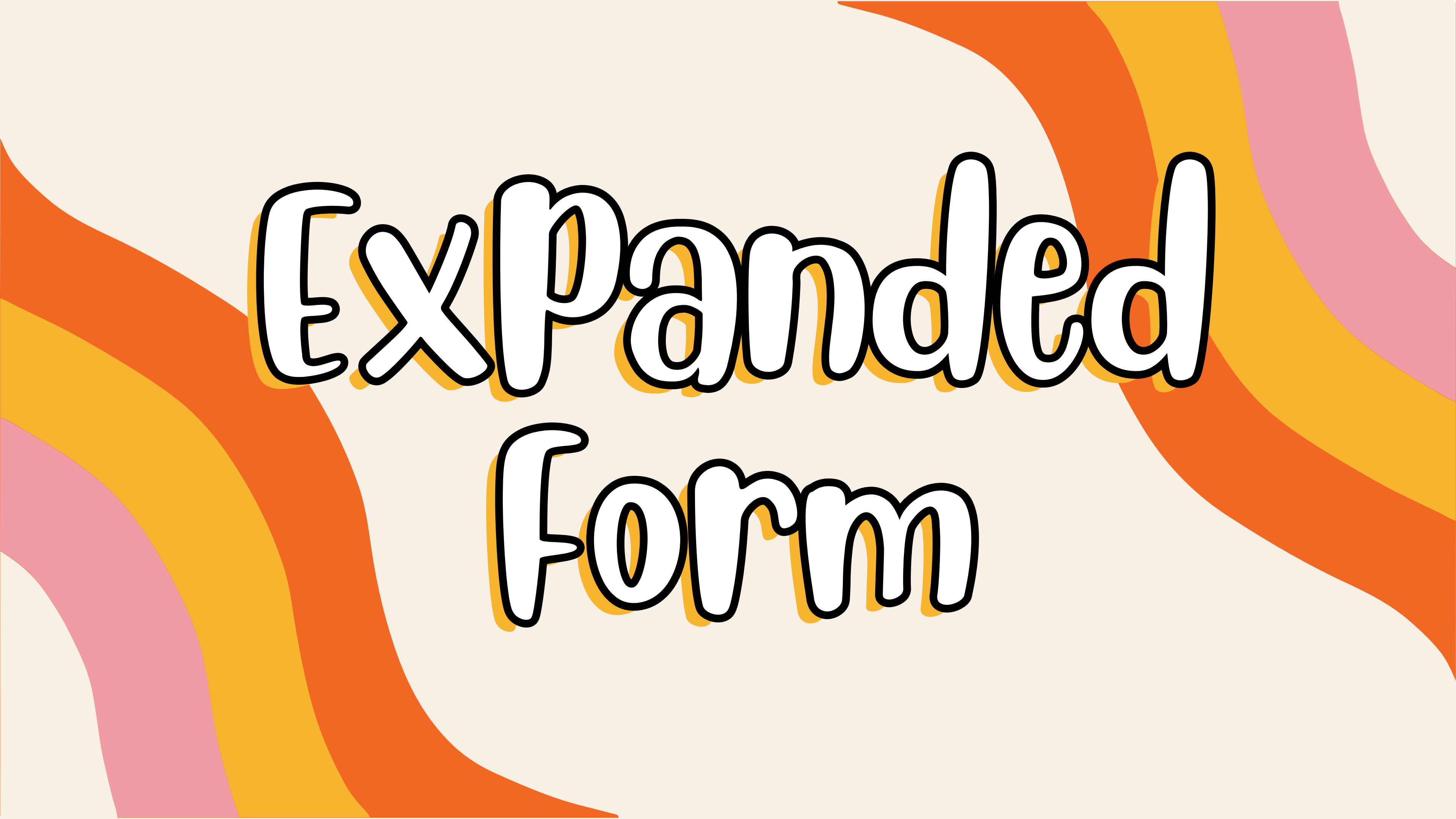
EXPANDED FORM

$2000 + 300 + 10 + 2$

BASE TEN BLOCKS



ANSWER



Expanded
Form

Expanded Form

Another name for expanded form is
expanded notation.

Expanded Form

To write a number in expanded form, we need to know the value of each digit.

Expanded Form

To write a number in expanded form, we need to know the value of each digit.

Expanded Form

Let's practice with the number 123.

The value of the 1 is 1 hundred, which is 100.

Expanded Form

The value of the 2 is 2 tens, which is 20.

The value of the 3 is 3 ones, which is 3.

Expanded Form

Write those values as a sum:

$$100 + 20 + 3$$

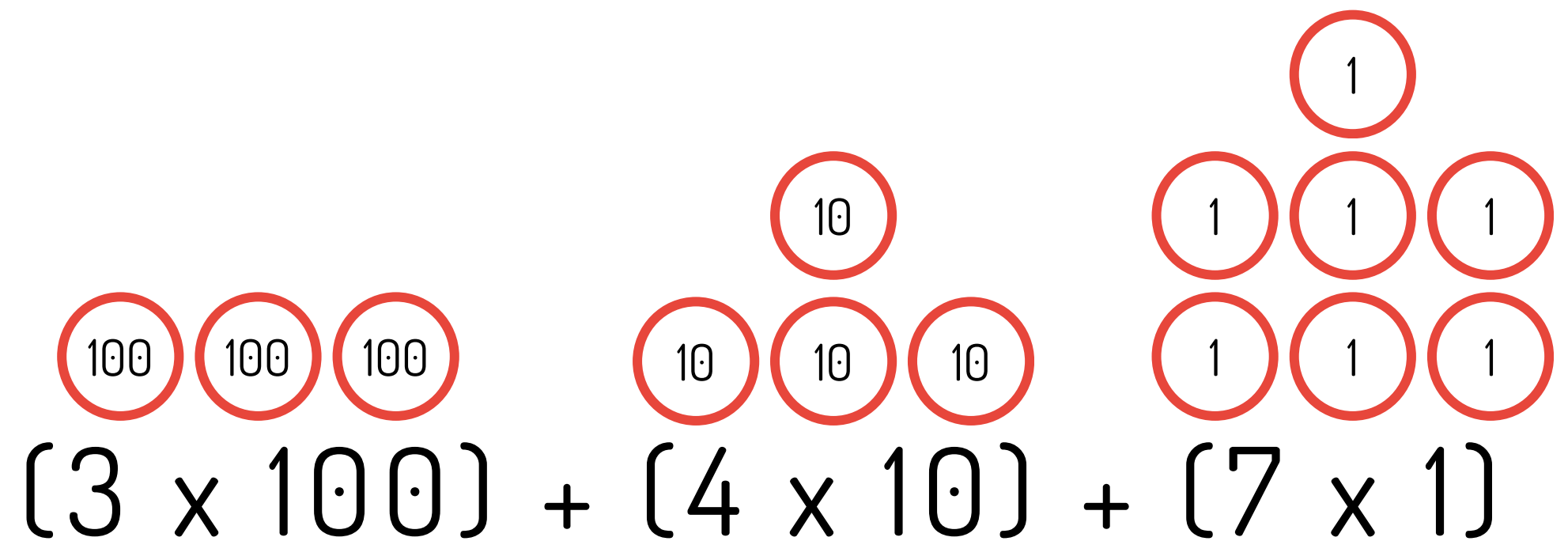
Expanded Form

So 123 in expanded form is:

$$100 + 20 + 3$$

EXPANDED NOTATION

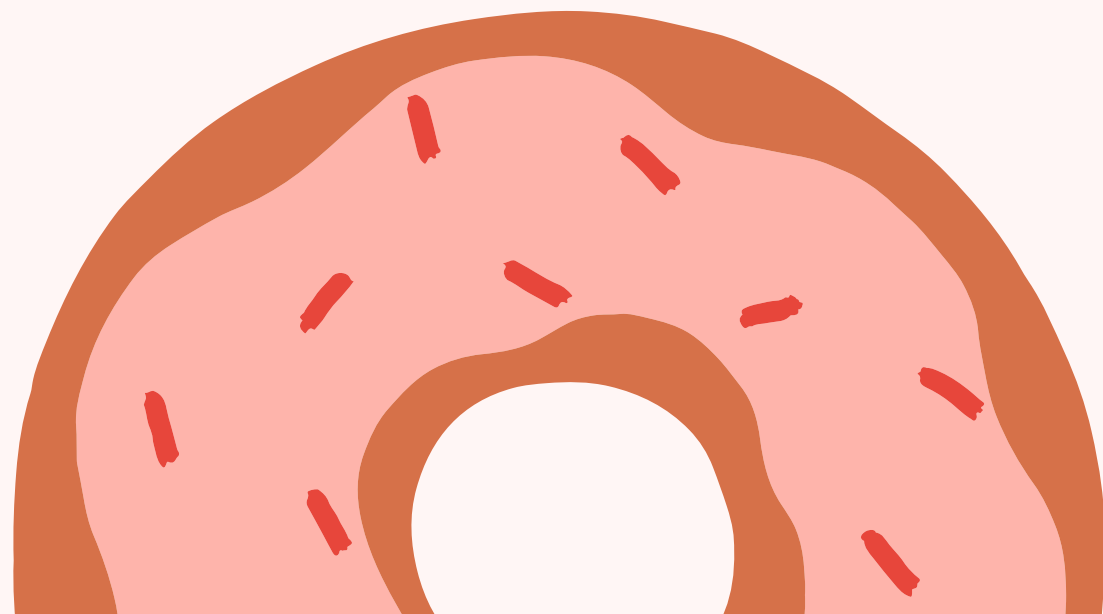
347



The diagram illustrates the expanded notation of the number 347. It features three groups of circles representing place values. The first group, for the hundreds place, consists of three circles, each containing the number 100. The second group, for the tens place, consists of four circles, each containing the number 10. The third group, for the ones place, consists of seven circles, each containing the number 1. Below these circles is the equation $(3 \times 100) + (4 \times 10) + (7 \times 1)$.

$$(3 \times 100) + (4 \times 10) + (7 \times 1)$$

represents each digit
multiplied by its place value.



Expanded Form

Time to practice! Write the numbers on the following slides in expanded form.

Expanded Form

362

Expanded Form

951

Expanded Form

5,329

estimation

GRADE 3



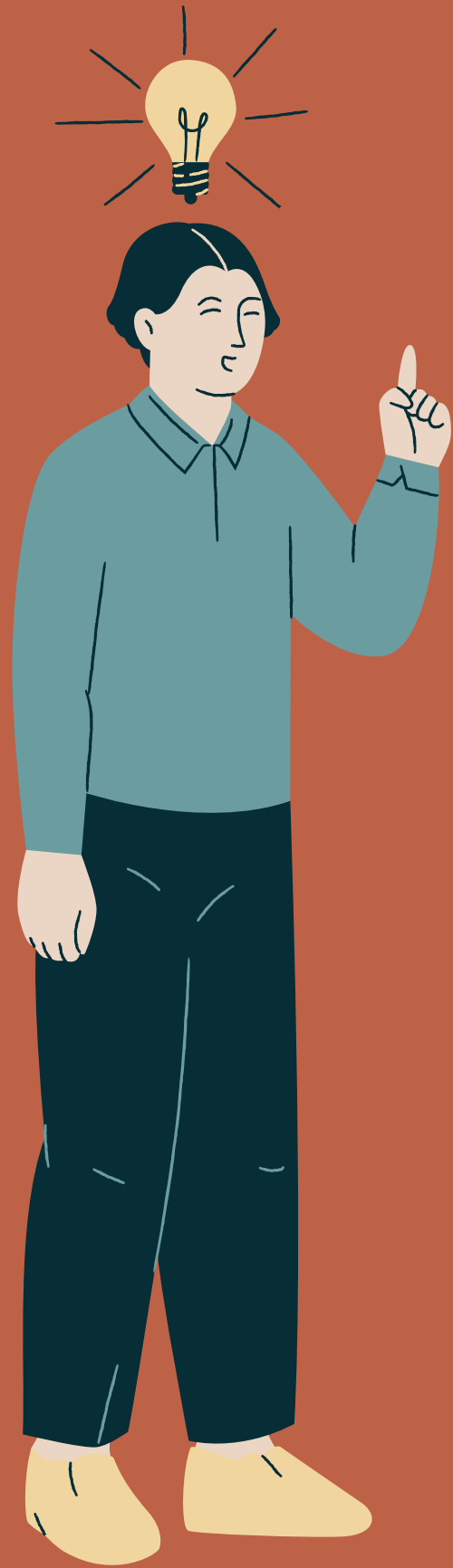
what is estimation?

An estimation is a rough calculation of something.



For example:

If we spent \$4.80 at the supermarket, we could say we spent about \$5.



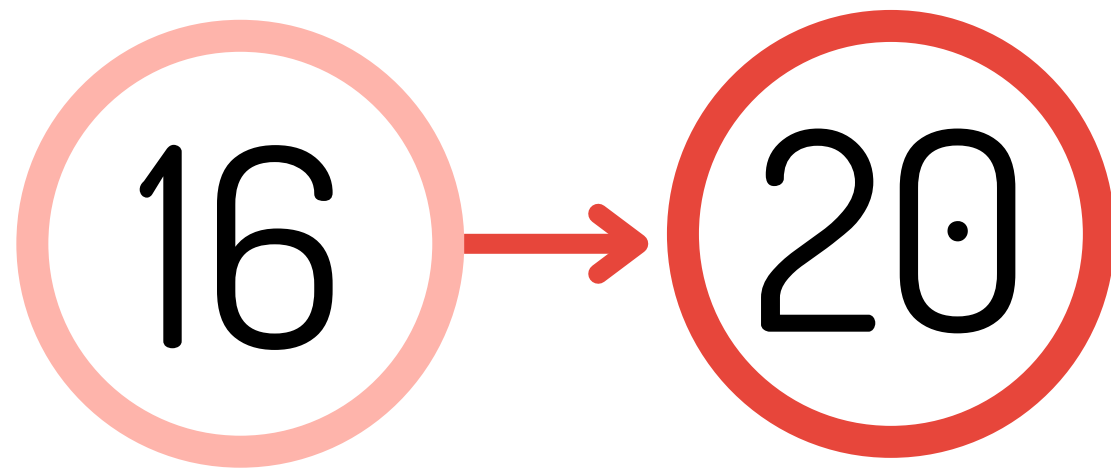
rounding

ROUNDING is a great tool to help us estimate.

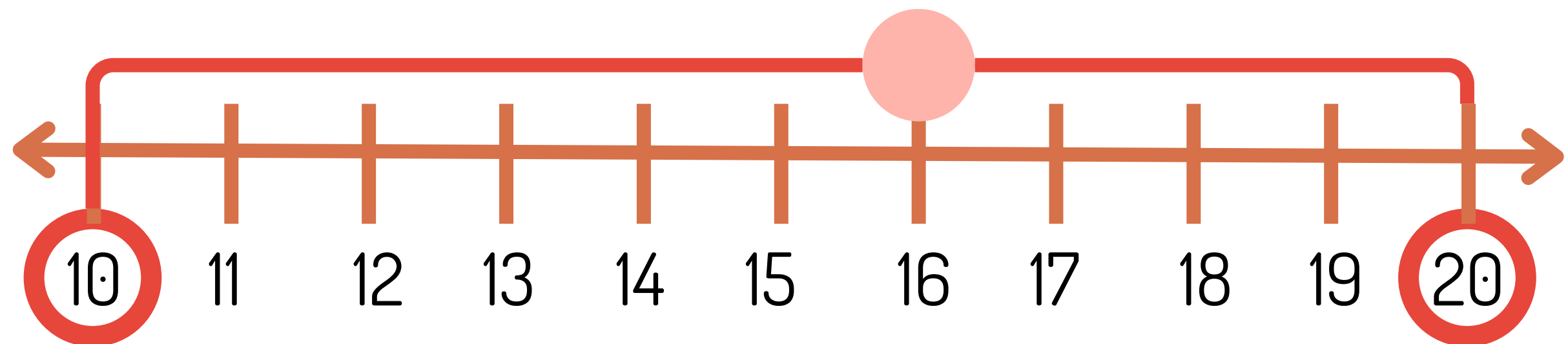
ROUNDING makes numbers easy to use in calculations but is less accurate.

Rounding

to the nearest ten

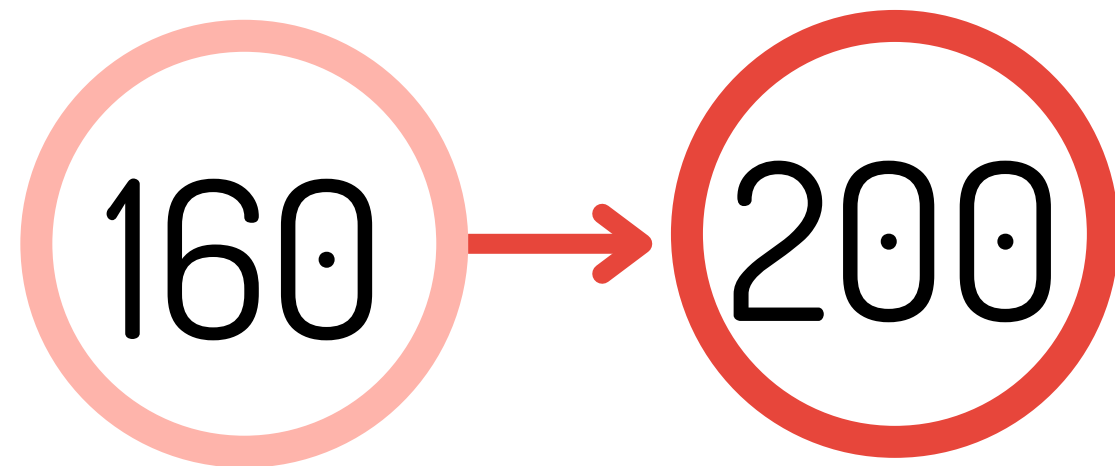


1. Find the number on the number line.
2. Determine the two boundary numbers.
3. Round to the number which is the closest.

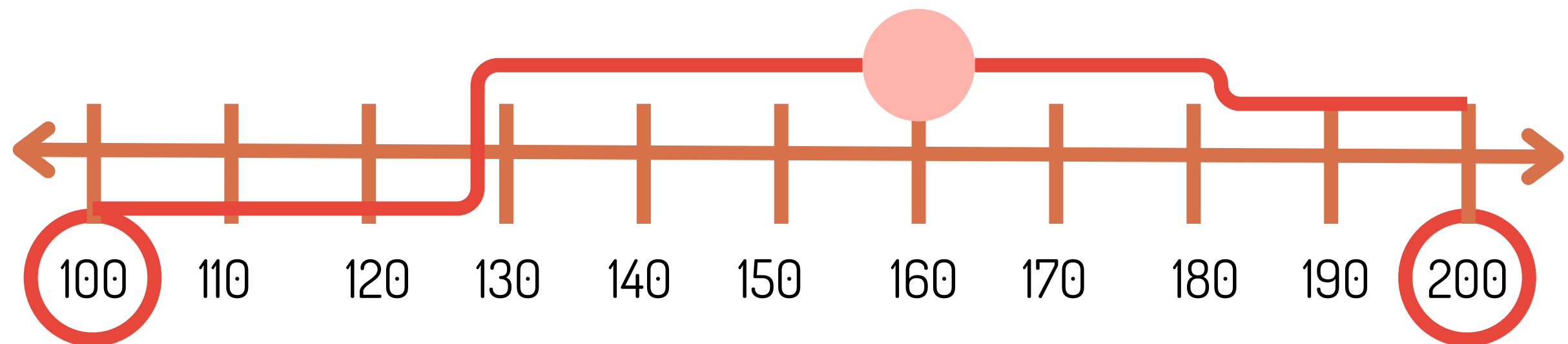


Rounding

to the nearest hundred



1. Find the number on the number line.
2. Determine the two boundary numbers.
3. Round to the number which is the closest.



tips for rounding



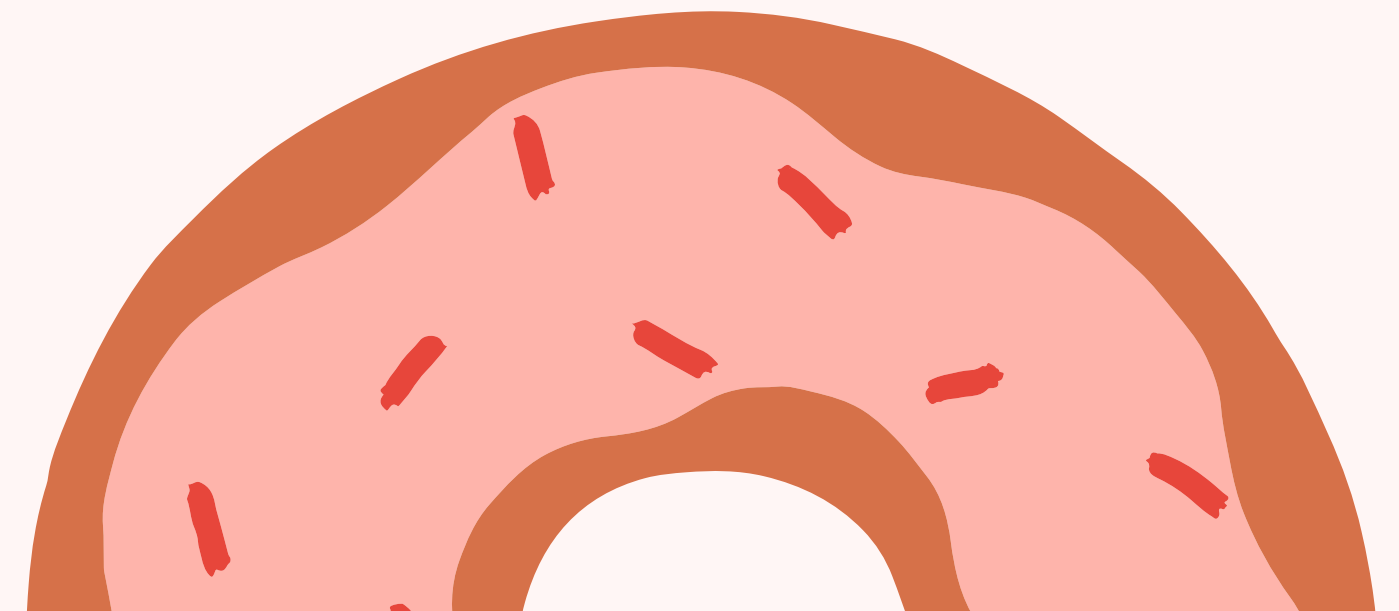
4 or less rounds DOWN

5 or more rounds UP



QUIZ TIME

LET'S PLAY A GAME



MULTIPLE CHOICE

WHAT IS THE SMALLEST WHOLE
NUMBER THAT WILL ROUND TO
50?

A 41

B 45

C 49

D 46

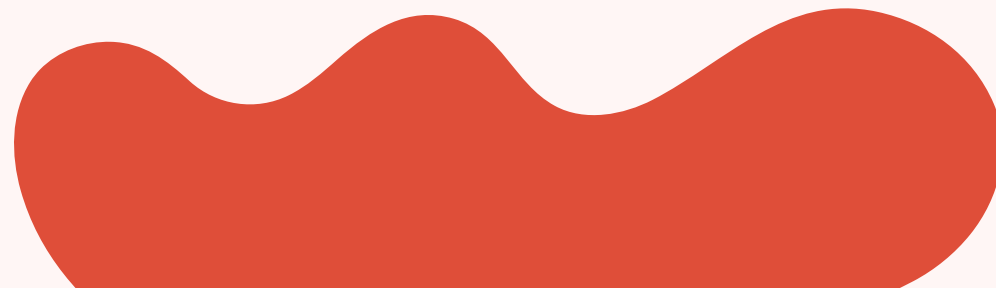


TRUE OR FALSE

549 IS THE LARGEST WHOLE
NUMBER THAT WILL ROUND TO
500.

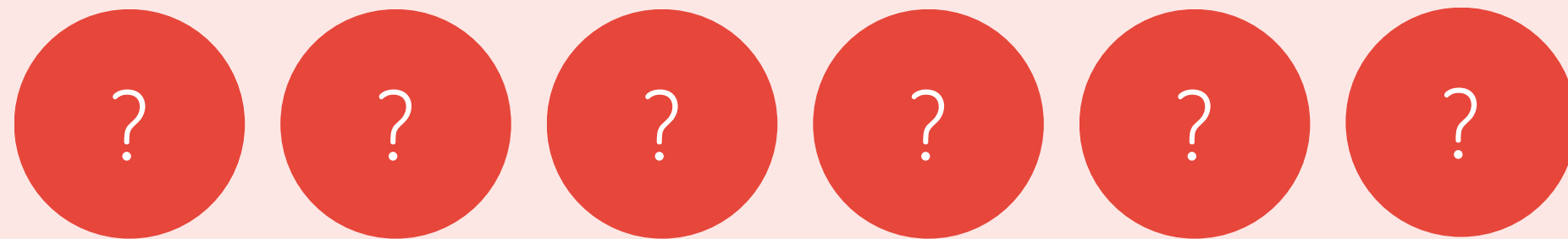
TRUE

FALSE



FILL IN THE BLANK

IN THE BASE-TEN SYSTEM,
EVERY PLACE GETS TEN TIMES



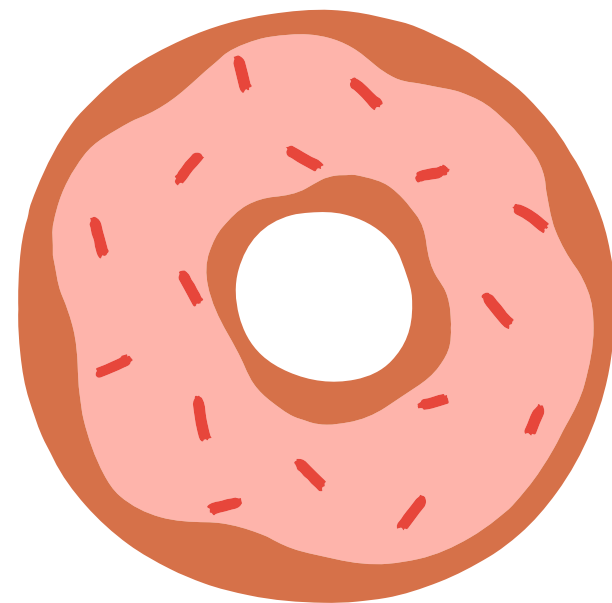
AS YOU MOVE TO THE RIGHT.

FIND THE ANSWER

ROUND TO DETERMINE THE
ANSWER:

$$764 + 29 =$$





ANSWERS

LET'S SEE HOW YOU DID!

MULTIPLE CHOICE

WHAT IS THE SMALLEST WHOLE
NUMBER THAT WILL ROUND TO
50?

A

41

C

49

B

45

D

46

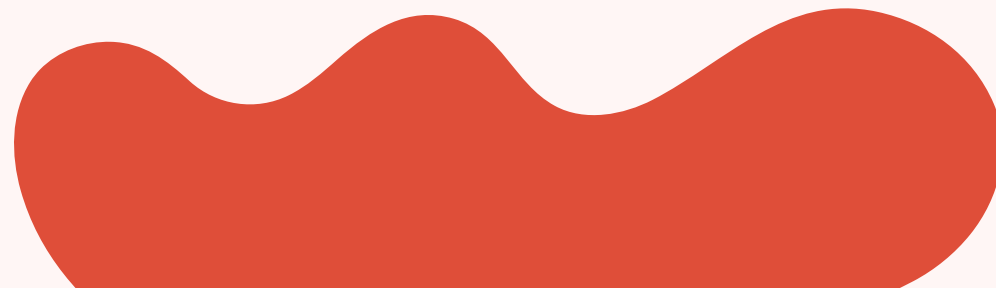


TRUE OR FALSE

549 IS THE LARGEST WHOLE
NUMBER THAT WILL ROUND TO
500.

TRUE

FALSE



FILL IN THE BLANK

IN THE BASE-TEN SYSTEM,
EVERY PLACE GETS TEN TIMES

B I G G E R

AS YOU MOVE TO THE RIGHT.



FIND THE ANSWER

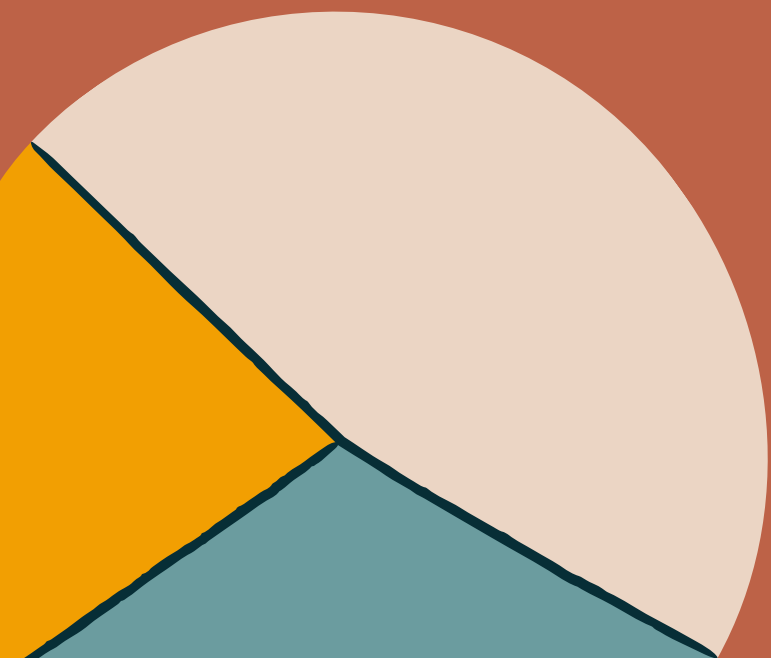
ROUND TO DETERMINE THE
ANSWER:

$$764 + 29 =$$

793



**let's try a few
other
questions...**



question 1.

A farmer estimates that there are 290 cows and 525 goats in a particular pasture. What is the total livestock population in the pasture? Choose the better estimate.

- A) 750 animals
- B) 1,000 animals
- C) 800 animals



question 2.

An ecologist believes that there are 4,256 oak trees and 8,975 pine trees in a designated forest area. What is the total tree population in the forest? Choose the better estimate.

- A) 14,000 trees
- B) 12,000 trees
- C) 13,000 trees



question 3.

A group of students were asked to choose Maths or Science as their favourite subject. Out of 240 students, 190 chose Maths, and 50 chose Science. Choose the best estimate.

- A) Maths: 70%, Science: 30%
- B) Maths: 80%, Science: 20%
- C) Maths: 60%, Science: 40%





1.

C

2.

C

3.

B



THANK
YOU

